

Concordance between NCO 2004 and NCO 2015 at the 3-digit group level

The National Classification of Occupations (NCO) is the official classification system for identifying occupations in Indian survey data. It is collated, developed and maintained by the Ministry of Labor and Employment, Government of India. The NCO organizes occupations in a tiered structure. At the highest level, it categorizes occupations into 9 broad divisions labelled by 1-digit codes and at the most granular level, occupations are identified by means of a 6- or 8- digit code supplemented with occupational titles. With the changing nature of occupations, the NCO has undergone a series of upgrades over the years, most recently when NCO 2004 was updated to NCO 2015. The Ministry of Labour and Employment makes available a concordance between NCO 2015 and NCO 2004 at the occupation level (6 or 8-digit) that depicts the changes made in the NCO code for each occupation.

The Periodic Labor Force Survey (PLFS), which is the primary source of employment data in India, codifies occupations of respondents at the group level (3-digit) across all years. Early rounds of PLFS (2017-18, 2018-19 and 2019-20) use the NCO 2004 while from 2021 onwards, the survey uses the NCO 2015. In order to conduct research on occupations over all the PLFS years, it is necessary to harmonize these two different classification systems. To that end, we overlaid the NCO 2015 codes, at the group (3-digit) level, onto the NCO 2004 codes to create a concordance between the two occupational classifications. This will enable users to identify occupation codes based on NCO 2015 from all years of the PLFS data, facilitating economic analysis and temporal comparisons of occupations.

Table 1: Components of a sample NCO code ‘1234.56’

Coding Structure NCO – 2015 Digits	Representation
The first digit	Division
The first two digits	Sub-Division
The first three digits	Group
The first four digits	Family
The first two digits after the decimal	Occupation

To create the 3-digit group concordance, we utilize the concordance between NCO 2015 and NCO 2004 at the occupation level provided by the Ministry of Labour and Employment. At the outset, in order to make this concordance compatible with the PLFS data, we collapse the 6 digit occupational codes to the 3 digit (group level) codes. As a result, we end up with a one-to-many

mapping between NCO 2004 and NCO 2015 for several groups. For example, 122_{nco04} mapped to (121, 131, 132, 133, 141, 142, 265)_{nco15}.¹ Due to reorganization of occupations between the two NCO classifications, a single 3-digit code in NCO 2004 can be replaced with several 3-digit codes in NCO 2015 as occupations (6-digit level) within a single NCO 2004 group were reclassified across several NCO 2015 groups. However, for our concordance to work within the PLFS dataframe, we needed each NCO 2004 group to match to only one NCO 2015 group (could be non-unique). In order to create the required concordance, we undertook the following steps.

I. Concordance at the three-digit level

A. Codes with a single match directly available in the concordance

Out of a total of 111 group level codes in NCO 2004, 50 codes had an exact, single match with a NCO 2015 group as per the official concordance document.² For each code with an exact match, we replaced the 2004 group code with the corresponding 2015 group code. In 11 out of these 50 cases, the codes did not change between the two classifications but the occupation titles did change for most (examples a, b, and c in Table 2). In the remaining 39 cases, codes changed between 2004 and 2015 and occupational titles for most codes also changed (examples d and e in Table 2).

Table 2: Examples of NCO04 codes with a single match in NCO15

Example	NCO 2004 code	Description	NCO 2015 code	Description
a.	931	Mining and Construction Labourers	931	Mining and Construction Labourers
b.	212	Mathematicians, Statisticians, and Related Professionals	212	Mathematicians, Actuaries, and Statisticians
c.	613	Market Oriented Crop and Animal Producers	613	Mixed Crop and Animal Workers
d.	233	Other Teaching Professionals	235	Other Teaching Professionals
e.	344	Customs, Tax, and Related Govt. Associate Professionals	355	Government Regulatory Associate Professionals

¹ Here onwards all NCO codes are subscripted with their respective year.

² There were a total of 116 groups in 2004. Out of these, we do not consider X00, X09, and X10. Codes 113 and 348 do not find a map to 2015 group level codes in the official concordance so we manually assigned 2015 group codes based on similarity with occupational titles (113_{nco04} was assigned to 111_{nco15} and 348_{nco04} was assigned to 343_{nco15}). As a result, we worked with 111 group codes from the official concordance.

All the NCO 2004 codes matched using this method are listed below:

111, 112, 114, 121, 212, 223, 231, 232, 233, 242, 243, 246, 312, 314, 323, 324, 331, 332, 333, 334, 342, 344, 345, 346, 412, 413, 422, 511, 515, 516, 521, 523, 612, 613, 614, 620, 712, 732, 733, 744, 817, 828, 831, 833, 834, 912, 920, 931, 932, 933

B. Overlap-based replacement

Next, we looked at how occupations (6-digit) within each of the remaining 61 unmatched NCO 2004 groups were reorganized in NCO 2015. The challenge was that for these 61 unmatched NCO 2004 groups, multiple 2015 groups could serve as the replacement. This is because occupations from each of the NCO 2004 groups were redistributed across several NCO 2015 groups during reclassification.

In the first step, we calculated the proportion of occupations that overlapped between a NCO 2004 group code and each of its 2015 counterparts. If 70 percent or more of the occupations listed under a group in 2004 were reclassified into a single 2015 group, then we use that particular 2015 group as a replacement. We use this as a threshold, i.e. 70 percent or more occupations became reclassified as part of a particular 2015 group but the other 30 percent or less were moved into other groups. To illustrate, 244_{nco04} represented ‘Social Science and Related Professionals’ and contained 49 occupations (6-digit) within its fold. 244_{nco04} maps to 263_{nco15} ‘Social and Religious Professionals’ (containing 48 occupations) and 264_{nco15} ‘Authors, Journalists, and Linguists’ (26 occupations). The overlap in occupations between 263_{nco15} and 244_{nco04} was 45 (such as Economist, Historian, and Biographer) and between 264_{nco15} and 244_{nco04} was 4 (such as Translator and Interpreter). In proportions, the overlap between the former set is 91.8 percent (45/49) and the latter set is 8.2 percent (4/49). Given the 70 percent threshold criteria, 244_{nco04} was replaced with 263_{nco15}. In essence, for cases of multiple mapping, we facilitate the assignment if a group satisfies the 70 percent likelihood criteria.

Using this approach, we could resolve another 32 codes out of the remaining 61. These are:

211, 214, 221, 244, 311, 411, 414, 421, 514, 522, 611, 615, 713, 714, 721, 722, 723, 731, 734, 741, 742, 811, 813, 814, 815, 816, 821, 823, 826, 827, 915, 916

Table 3: Matches made with the threshold method

nco_2004	total_nco04_obs	mapping_list
211	34	211 (N=33, 97.1%) 226 (N=1, 2.9%)
214	102	214 (N=77, 75.5%) 215 (N=14, 13.7%) 216 (N=11, 10.8%)

221	38	213 (N=31, 81.6%) 221 (N=6, 15.8%) 225 (N=1, 2.6%)
244	49	263 (N=45, 91.8%) 264 (N=4, 8.2%)
311	62	311 (N=58, 93.5%) 352 (N=4, 6.5%)
411	18	412 (N=4, 22.2%) 413 (N=14, 77.8%)
414	18	334 (N=5, 27.8%) 441 (N=13, 72.2%)
421	16	421 (N=13, 81.2%) 422 (N=3, 18.8%)
514	15	514 (N=11, 73.3%) 516 (N=4, 26.7%)
522	14	522 (N=3, 21.4%) 524 (N=11, 78.6%)
611	19	611 (N=18, 94.7%) 613 (N=1, 5.3%)
615	23	622 (N=18, 78.3%) 754 (N=5, 21.7%)
713	37	312 (N=4, 10.8%) 712 (N=27, 73.0%) 741 (N=6, 16.2%)
714	16	312 (N=2, 12.5%) 713 (N=14, 87.5%)
721	40	312 (N=2, 5.0%) 721 (N=35, 87.5%) 754 (N=3, 7.5%)
722	36	312 (N=2, 5.6%) 722 (N=34, 94.4%)
723	45	312 (N=3, 6.7%) 712 (N=1, 2.2%) 723 (N=41, 91.1%)
731	53	312 (N=3, 5.7%) 321 (N=1, 1.9%) 731 (N=48, 90.6%) 818 (N=1, 1.9%)
734	45	312 (N=2, 4.4%) 732 (N=37, 82.2%) 813 (N=6, 13.3%)
741	57	312 (N=4, 7.0%) 751 (N=53, 93.0%)
742	53	312 (N=1, 1.9%) 731 (N=13, 24.5%) 752 (N=39, 73.6%)
811	35	312 (N=9, 25.7%) 811 (N=26, 74.3%)
813	34	312 (N=2, 5.9%) 818 (N=32, 94.1%)
814	38	312 (N=2, 5.3%) 313 (N=5, 13.2%) 817 (N=31, 81.6%)

815	73	312 (N=2, 2.7%) 313 (N=17, 23.3%) 813 (N=54, 74.0%)
816	34	312 (N=2, 5.9%) 313 (N=24, 70.6%) 818 (N=8, 23.5%)
821	36	312 (N=2, 5.6%) 722 (N=27, 75.0%) 811 (N=7, 19.4%)
823	44	312 (N=2, 4.5%) 814 (N=42, 95.5%)
826	118	312 (N=6, 5.1%) 815 (N=112, 94.9%)
827	83	312 (N=2, 2.4%) 816 (N=81, 97.6%)
915	17	541 (N=2, 11.8%) 962 (N=15, 88.2%)
916	7	961 (N=6, 85.7%) 962 (N=1, 14.3%)

C. Coalescing 3-digit groups

29 NCO 2004 codes could not be resolved using the overlap-based replacement strategy as none of the counterparts in NCO 2015 met the 70 percent threshold. For these remaining 29 codes, we manually created new groups aggregating and coalescing wherever necessary. In practice, we congregated similar groups among the 29 codes in NCO 2004, and their NCO 2015 counterpart codes, into new categories while preserving their original occupational identity as much as possible. New occupational titles were also created for the new categories. This coalesced similar occupational groups together since the tasks and responsibilities as described in the occupational descriptions were closely aligned.

Throughout this aggregation/reorganization exercise, we based our decisions on some common principles which were derived after a careful scrutiny of occupational descriptions. One principle was to preserve occupational (or skill) hierarchies as much as possible. The 1-digit aggregation level in the NCO 2015, called division, reflects a hierarchy of skills, with division 1 representing the highest level of skill and division 9 representing the lowest skill level. This meant that we resisted consolidation of groups from disparate 1-digit divisions as much as possible. For example, (121, 122, 131, 132, 133, 134, 141, 142, 143)_{nco15} was coalesced together into a new category which represented 'Managers across Industries' with their 2004 equivalent (122, 123, 130) representing the same new category where the division was 1 throughout.

As another example, consider the following three groups 512_{nco04} (House Keeping and Restaurant Service Workers), 913_{nco04} (Domestic and Related Helpers, Cleaners, and Launderers) and 914_{nco04} (Building Caretakers, Window, and Related Cleaners). 913_{nco04} maps to 911_{nco15} (Domestic,

Hotel, and Office Cleaners and Helpers), 912_{nco15} (Vehicle, Window, Laundry, and Other Hand Cleaners), and 941_{nco15} (Food Preparation Assistants). 914_{nco04} maps to 912_{nco15} and 515_{nco15} (Building and Housekeeping Supervisors). We reasoned that since both 913_{nco04} and 914_{nco04} fall within the same 1-digit division called Elementary Occupations, it is most reasonable to aggregate them together which would preserve their occupational hierarchy and also keep intact their mapping to elementary occupations in NCO 2015 such as those that begin with a 9 like 911_{nco15}, 912_{nco15}, and 941_{nco15}. Additionally, these three groups in NCO 2015 were exclusive to 913_{nco04} and 914_{nco04} so their aggregation would not pose any compositional change in the final concordance. Lastly, 515_{nco15} also mapped to 512_{nco04} and it seemed more accurate to place “supervisors” with “service workers” instead of elementary occupation workers. Therefore, 515_{nco15} was retained in 512_{nco04}. Finally, as a result, we were able to get a unique new category comprising 913_{nco04} and 914_{nco04} with the corresponding groups 911, 912, and 941 in 2015. This new category was numbered and defined singularly too.³

Lastly, for all the newly created categories, we generally followed the naming convention of concatenating the titles of the constituent groups from NCO 2015. The newly created categories, their codes, and their equivalent 2004 and 2015 codes are shown in the table below. These newly created codes can now be inserted into the NCO 2004 and 2015 classifications to replace the original codes and descriptions.

nco_2004_code	nco_2004_code_description	new_category	new_code	includes_2015_codes
122, 123, 130	Production and Operations Department Managers, Other Department Managers, General Managers	Managers (Across Industries)	1_n	121, 122, 131, 132, 133, 134, 141, 142, 143
213	Computing Professionals	Software, Database and Network Professionals	2_n1	251, 252
222	Health Professionals (except nursing)	Doctors, Veterinarians and Complementary Medicine (Except Nursing)	2_n2	221, 223, 225
241	Business Professionals	Business, Financial, and Sales Professionals	2_n3	241, 242, 243
245	Writers or Creative and Performing Artists	Architects, Writers, and Creative Artists	2_n4	216, 264, 265
313, 315	Optical and Electronic Equipment Operators + Safety and Quality Inspectors	Science and Telecommunications Technicians and Other Craft Workers	3_n1	311, 352, 754
321	Life Science Technicians and Related Health Associate Professionals	Life Science Professionals and Associate Professionals	3_n2	213, 314
322	Modern Health Associate Professionals (Except Nursing)	Medical, Paramedical, Pharmaceutical Technicians and Others (Except Nu	3_n3	224, 226, 321, 324, 325
341	Finance and Sales Associate Professionals	Business, Financial, and Sales Associate Professionals	3_n4	331, 332, 333
343, 348	Administrative Associate Professionals	Administrative, Legal, Social, Religious, Associate Professionals	3_n5	334, 341
347	Artistic, Entertainment and Sports Associate Professionals	Sport and Entertainment Workers and Administrative Associate Professio	3_n6	342, 343
419	Other Office Clerks	Office Clerks and Clerical Support Workers	4_n1	411, 441
512	Housekeeping and Restaurant Service Workers	Cooks, Serving Staff, and Housekeeping Supervisors	5_n1	512, 513, 515
513	Personal Care and Related Workers	Childcare and Personal Care Workers	5_n2	531, 532
711	Miners, Shotfirers, Stone Cutters and Carvers	Building Frames and Related Workers and Plant Operators (Mining and N	8_n1	711, 811
724	Electrical and Electronic Equipment Mechanic and Fitters	Installers and Repairers (Electrical, Electronics, and Telecommunications	7_n1	741, 742
812	Metal Processing Plant Operators	Supervisors and Technicians (Metal Processing, Mining, Manufacturing, Construction, and Process Control)	3_n7	312, 313
822, 825, 824, 829, 743	Chemical Products Machine Operators, Printing, Binding and Paper Products Machine Operators, Textile, Garment and Related Trades Workers + Wood Products Machine Operators + Other Machine Operators and Assemblers	Workers and Plant Operators (Handicraft, Garment, Wood Treaters, Printing Trades, Metal Processing, Chemicals, and Textile)	8_n2	731, 732, 752, 753, 812, 813, 814, 815, 818
832	Motor Vehicle Drivers	Vehicle Drivers (Car, Van, Motorcycle, Trucks, and Buses)	8_n3	832, 833
911	Street Vendors and Related Workers	Street and Market Sales workers and Street Vendors (Excluding Food)	5_n3	521, 524, 952
913, 914	Domestic and Related Helpers, Cleaners and Launderers + Building Caretakers, Window Cleaners, Helpers, and Food Preparation Assistants		9_n	911, 912, 941

³ Figure 1 explains this example diagrammatically.

II. Concordance at the 1-digit level

For aggregating to the 1-digit (division) level, we needed to determine which division the newly created categories would be located in. This was needed to enable occupational analysis of PLFS data at the 1-digit level. To this end, we used a simple majority rule. If a majority of the constituent groups that make up a new group belonged to a certain division, the category was assigned that division. Correspondingly, the codification process reflected this mechanism by incorporating that digit at the beginning, suffixed by a ‘_n’ or ‘_n1’, ‘_n2’. For example, “Street and Market Sales workers and Street Vendors (Excluding Food)” were assigned division 5 with a code ‘5_n3’. 5_n3 is a new group which contains 521_{nco15}, 524_{nco15}, and 952_{nco15}. Since 2 of these groups belong to division 5 and 1 group belongs to division 9, by the simple rule of majority, the whole group was assigned to division 5 and codified with a ‘5’ in the beginning.

In cases where the majority rule did not deliver a clear answer, we looked at occupations (6-digit) to determine the majority. For instance, “Life Science Professionals and Associate Professionals” is made up of 314_{nco15} and 213_{nco15}. It was assigned division 3 because there were three 6-digit occupations within 314 and only one such occupation within 213. Similarly, “Workers and Plant Operators (Handicraft, Garment, Wood Treaters, Printing Trades, Metal Processing, Chemicals, and Textile)” was assigned division 8 as there were 25 occupations with 811 and only 5 occupations with 711.

Limitations of this approach

While the concordance is necessary and makes possible over-time analysis of occupations using the PLFS, there are limitations to it. First, while applying the threshold method, we had to assign a 2015 group to a 2004 group based on a 70 percent threshold even if it practically meant reclassifying the remaining 30 percent occupations to that group too. For example, 414_{nco04} mapped to 334_{nco15} and 441_{nco15}. In terms of overlapping proportions of occupations, the overlap between 414_{nco04} and 334_{nco15} was 27.8 percent and between 414_{nco04} and 441_{nco15} was 72.2 percent. As a result of the threshold criteria of 70 percent, 414_{nco04} was mapped to 441_{nco15}. While cases of a near 30 percent rearrangement are few, any threshold-based criteria will result in some magnitude of rearrangement. Given the lack of more granular data regarding a worker’s occupation in the PLFS, we found this to be the only meaningful way to assign the group among the two options in NCO 2015.

Secondly, as discussed above, one of the constraints in building the concordance pertained to restricting 2015 codes to one category despite them mapping onto more than one category. This also resulted in some compositional rearrangement.

Third, we developed the concordance at the 3-digit level and then aggregated it to 1-digit level. We did not produce a corresponding sub-division (2-digit) level crosswalk.

Fourth, the structural misalignment between NCO 2004 and NCO 2015 at the 3-digit level necessitated bypassing the occupational hierarchy principle in some instances. In order to operationalize the concordance for PLFS—which requires a single unique identifier per worker—forced one-to-one mappings were required for groups that naturally span multiple categories. For example, 265_{nco15} spans 122, 245, and 347 in NCO 2004 each of which coalesce into distinct new categories. However, 265_{nco15} had to be restricted to the category where 245 is included. At other times, to prevent distorting the compositional mix to a large extent, codes at different levels of hierarchy were coalesced together to maintain consistency for individual occupations that were redistributed over divergent groups between 2004 and 2015.

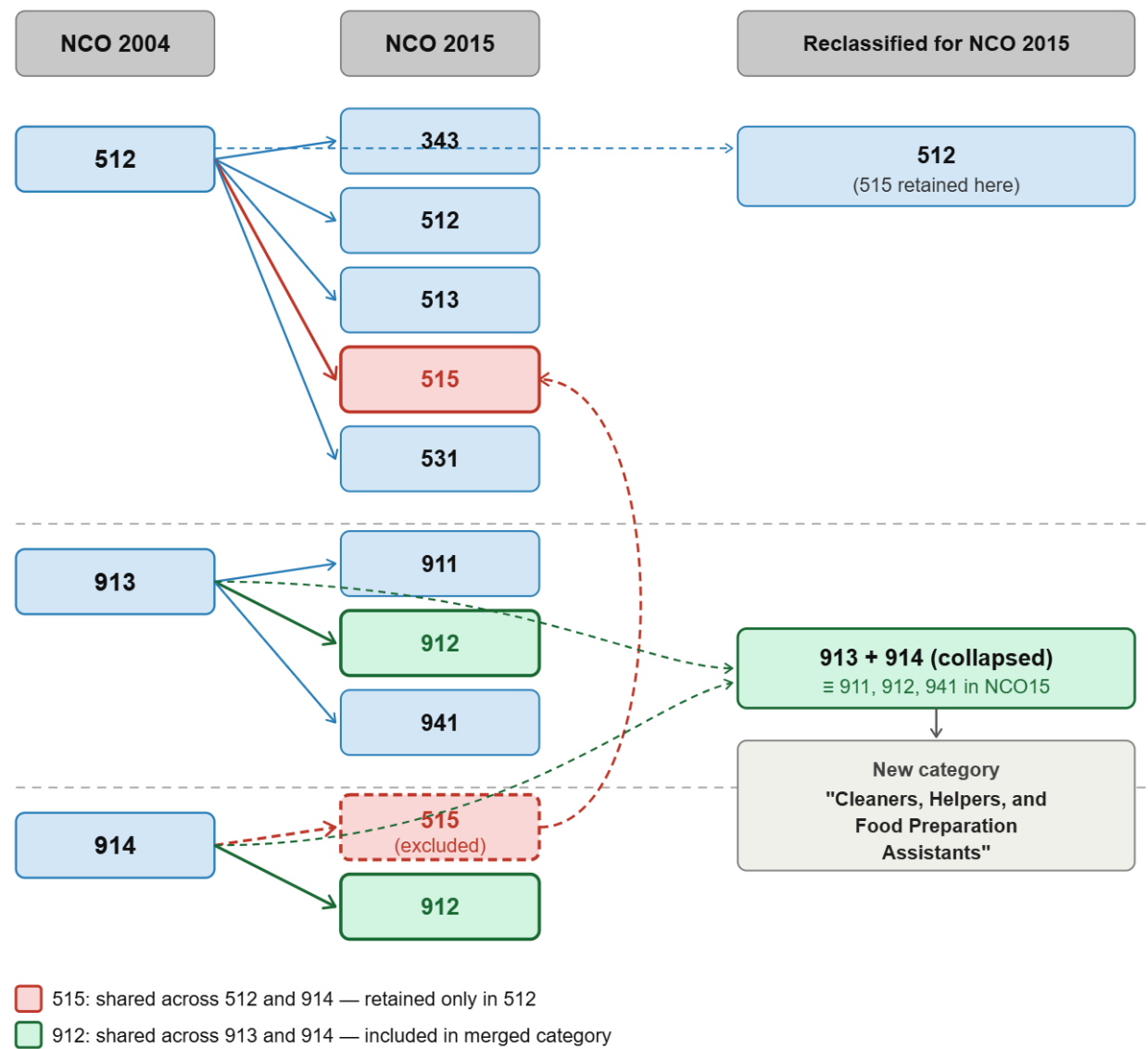
Final concordance

Due to the aggregation exercise, the final concordance resulted in a shrinking of the number of groups at the 3-digit. In the original NCO 2015 classification system, there were 127 groups. After applying the frameworks outlined in this document, we get 81 groups. The citation for the concordance is provided below:

Ahmed, A., Mahajan, K., Sharma, A., & Wadhwa, M. (2026). Group level concordance between NCO-04 and NCO-15 (Version 1.0.0) [Data set].

<https://github.com/India-NCO-04-NCO-15-3-digit-concordance/concordance>

Figure 1: Illustration of the example for coalescing exercise



Important Links

1. [NCO 2004 document](#)
2. [NCO 2015 document \(Vol I\)](#)